Sami Bekuma

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Professional Summary

Machine Learning Engineer with experience building and deploying production ML and LLM systems serving more than 20 million daily user interactions. Specialized in NLP, RAG pipelines, and intent classification, with hands-on expertise in cost optimization using Azure OpenAI and open-source models. Focused on improving model reliability and accelerating deployment cycles for high-traffic platforms through automated evaluation, efficient routing strategies, and MLOps best practices.

PROFESSIONAL EXPERIENCE

Machine learning Engineer
Comcast - Washington, DC 2024 –present

- Scaled the **Xfinity Voice Remote** AI platform to handle 20M+ daily voice commands across North America and Europe by architecting multi-language pipelines that translate speech into logical intents for device control (navigation, app launching)
- Reduced infrastructure costs by developing and productionizing a supervised intent-classification model, enabling real-time, cost-efficient routing between high-cost RAG/LLM pipelines and legacy CPU-based services, without compromising quality.
- Designed and deployed a latency-based load balancing strategy for production voice services, reducing request timeouts by 9%, supported by real-time Grafana observability for latency, throughput, and model performance.
- Accelerated release validation cycles by 85% by building an automated evaluation framework using Azure and fined tuned Qwen model as a judge within CI/CD, introducing scalable metrics for factuality, hallucination, relevance, and toxicity.
- Delivered key user-facing features for major live events, including content discovery for Olympics and NFL Football on Peacock by implementing NLP and business-logic enhancements, ensuring reliable content retrieval during high-traffic.

Lead-tutor
Tutoring Club of McLean, *McLean, VA* 2022 – 2024

- Tutored students **Java** and **Python programming**, and test preparation (SAT, ACT).

SKILLS

Languages & Frameworks: Python, Java, SQL, JavaScript, Spring Boot, React
Data & ML: Apache Airflow, Transformers (Hugging Face), LlamaIndex, Scikit-learn, Pandas, NumPy, LangChain, SentenceTransformers, SpaCy, NLTK, MLflow, VLLM, KServe, RAG System, Model Evaluation & Benchmarking, LLM-as-a-Judge
DevOps & Cloud Tools: AWS (SageMaker, EC2, S3, Lambda), Azure, Docker, Kubernetes, Git, Jenkins, Github Actions, Grafana
Soft Skills: Leadership, problem-solving, data quality mindset, cross-functional collaboration, strong communication

TECHNICAL PROJECTS

AI-Powered Doctor Appointment Booking System (React, Node.js, Express, MySQL, Google Gemini)

- Designed and implemented a **production-style healthcare scheduling platform** that replaces manual and phone-based appointment booking with a structured, self-service workflow.
- Integrated an **AI-powered chatbot (Google Gemini)** that interacts with backend APIs and the database to **create and cancel appointments in real time**, in addition to navigation and user assistance.
- Designed **backend scheduling and conflict-prevention logic** using Node.js, Express, and relational data modeling in MySQL.

Automated Real-Time Data Ingestion Pipeline with AWS and OpenSearch

- Designed and deployed an end-to-end cloud data ingestion pipeline using **AWS** services (**S3**, **SQS**, **Lambda**, **OpenSearch Service**) to automate real-time data processing and analytics.
- Configured an **event-driven architecture** to trigger processing on new **S3** uploads, enabling immediate parsing and indexing into **OpenSearch** for scalable search and visualization.

EDUCATION

GEORGE MASON UNIVERSITY, The Volgenau School of Engineering, Fairfax, VA Dec 2023
BS in Computer Science Grade: 3.64
Honors & Awards: 2022 All-Virginia Academic Team Award, Dean's List, Presidential Scholar